CLM-Net: A Channel-Optimized Lightweight Hybrid Network for Varroa Mite Segmentation

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Abstract. Accurate segmentation of Varroa destructor mites is important for automatic honeybee health monitoring, yet it remains challenging because mites occupy only a very small portion of the image and often appear with low contrast against the bee body. Although hybrid CNN–Transformer architectures provide a promising balance between local feature extraction and global contextual modeling, applying attention over all feature channels may introduce redundant feature interactions and unnecessary computational overhead. In this paper, we propose CLM-Net, a channel-optimized lightweight hybrid network for Varroa mite segmentation. CLM-Net is developed from LM-Net by introducing two lightweight components: the Lightweight Global Bottleneck (LGB), which compresses bottleneck features before global self-attention, and the Lightweight Neighborhood Attention Block (LNAB), which applies neighborhood attention only to partial feature channels in the skip pathway. These designs reduce redundant channel interactions while preserving global context and local structural details. Experiments on the Bee Varroa segmentation dataset show that CLM-Net-192 achieves the best Dice score of 70.32% and IoU of 54.24%, outperforming the original LM-Net while reducing the parameter count from 4.15M to 1.71M and GFLOPs from 9.899 to 8.005. These results demonstrate that channel-wise optimization is an effective strategy for lightweight small-object segmentation in precision apiculture.

Keywords: Varroa mite segmentation · small-object segmentation · lightweight semantic segmentation · channel reduction · feature projection

1 Introduction

Image semantic segmentation has emerged as a foundational task in computer vision, aimed at assigning precise pixel-level labels in images. While this technology has matured in domains like autonomous driving and general medical

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imaging, its application to agricultural challenges like Varroa destructor mite segmentation remains underexplored. Varroa mites are destructive parasites that devastate honeybee colonies so automated yet accurate segmentation is critical for early infestation assessment.

Convolutional neural networks (CNNs), including FCN-based and encoder–decoder architectures [1, 2, 3, 4], are effective at extracting local visual patterns but have limited capability for modeling long-range dependencies. Transformer-based models [5, 6, 7, 8] address this limitation through self-attention, although their computational cost can be substantial. Consequently, hybrid CNN–Transformer architectures [9, 10, 11] have emerged to combine local feature extraction with global contextual modeling.

However, many hybrid architectures apply attention across the full feature dimensionality, resulting in redundant computation for lightweight segmentation tasks involving very small targets. LM-Net [12], for example, employs attention at both the bottleneck and skip pathways. While effective for medical image segmentation, this design is less efficient for Varroa mite images, where the target occupies small portion of the image and most feature responses are dominated by background regions.

To address this issue, we develop CLM-Net, a channel-optimized lightweight extension of LM-Net for small-target segmentation. Instead of applying attention uniformly to all feature channels, CLM-Net compresses bottleneck features before global self-attention and applies neighborhood attention to only part of the skip-feature channels. The main contributions are summarized as follows:

- We introduce a lightweight global bottleneck that replaces the original patch embedding with a 1×1 channel projection followed by depthwise convolution, reducing attention cost while preserving spatial information for small targets.
- We redesign the original neighborhood-attention skip pathway by applying neighborhood attention to only a subset of feature channels while preserving the remaining channels through an identity pathway. This design reduces attention overhead while maintaining information for accurate mite segmentation.
- We demonstrate through comprehensive experiments that the proposed channel-optimized modifications reduce the parameter count from 4.15M to 1.71M (a 58.8% reduction) while improving the Dice score from 69.41% to 70.32% on the Varroa mite dataset.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the materials and methods, including the dataset and the proposed CLM-Net architecture. Section 4 describes the experimental setup and discusses the results. Finally, Section 5 concludes the paper and outlines future work.

2 Related Works

Varroa mite detection and segmentation have received increasing attention due to the impact of *Varroa destructor* on honeybee colony health. Early computer-vision studies mainly addressed detection and infestation monitoring. Bjerge et al. developed a portable vision system for estimating infestation levels from honeybee videos [13], while Bilik et al. evaluated YOLO- and SSD-based detectors for visual Varroa diagnosis [14]. These studies demonstrated the feasibility of automated monitoring but were limited to image-level assessment or bounding-box localization.

Later works introduced segmentation to improve localization and reduce background interference. Liu et al. combined bee-region segmentation with a YOLOX detector [15], Kriouile et al. modeled bee–Varroa detection as a nested-object problem using Mask R-CNN [16], and Bielik and Bilik explored U-Net-based segmentation under narrow-spectrum illumination [17]. However, accurate pixel-level segmentation from ordinary RGB bee images remains challenging.

The most closely related study is the benchmark by Ho et al. [18], which compared U-Net variants, DeepLab models, and YOLO-based methods on a Varroa segmentation dataset. DeepLabV3 with a ResNet101 backbone achieved the best reported result, with 52.11% IoU and 67.21% Dice. Nevertheless, the study mainly evaluated existing architectures rather than developing a model specifically for tiny and low-contrast mites.

Varroa segmentation is closely related to small-object segmentation because the mite occupies only a small image region and may be suppressed by repeated downsampling. Its appearance can also resemble the bee body, leading to severe foreground–background imbalance and boundary ambiguity. Previous studies have therefore emphasized high-resolution feature preservation, multi-scale representation, contextual modeling, and attention mechanisms [19, 20, 21].

LM-Net combines convolutional feature extraction with transformer-based contextual modeling in a lightweight multi-scale architecture [12]. However, it was not designed specifically for tiny and low-contrast Varroa mites. Our work therefore adapts LM-Net through channel-wise optimization to improve feature efficiency while maintaining a lightweight architecture.

3 Material and Method

3.1 Dataset

The Bee Varroa segmentation dataset introduced by Schurischuster and Kampel [22] was used in this study. The dataset was collected from video recordings captured at bee hives. The provided bounding-box annotations were converted into binary segmentation masks, where the foreground corresponds to Varroa mites and the background represents non-Varroa regions.

Table 1: Distribution of images and Varroa annotations across the training, validation, and testing sets

Split	**Healthy**	**Infested**	**Images**	**Varroa annotations**
Training	5,671	2,554	8,225	2,905
Validation	1,425	451	1,876	655
Testing	2,466	942	3,408	1,068
Overall	**9,562**	**3,947**	**13,509**	**4,628**
Resolution	160 × 280 pixels			

The dataset comprises 13,509 images with a fixed resolution of 160 × 280 pixels and is divided into 8,225 training images, 1,876 validation images, and 3,408 testing images, including 3,947 infested and 9,562 healthy bee images. The statistics are summarized in Table 1 and representative examples of healthy and Varroa-infested bee images together with their corresponding segmentation masks are shown in Fig. 1.

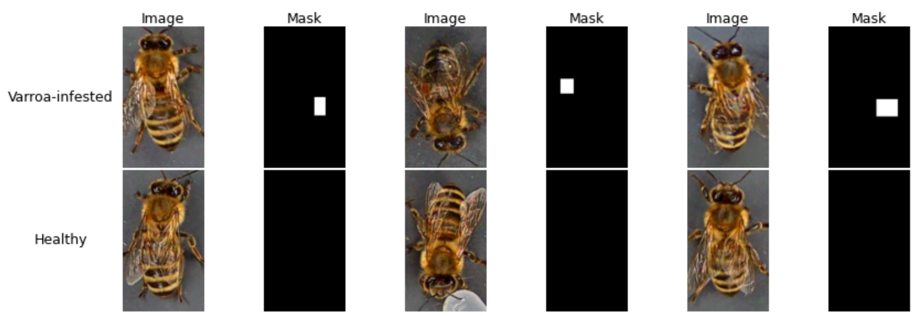

Fig. 1: Representative Bee Varroa images and corresponding segmentation masks

3.2 The Proposed CLM-Net Architecture

CLM-Net is a lightweight CNN–Transformer hybrid architecture developed from LM-Net for Varroa mite segmentation. As illustrated in Fig. 2, CLM-Net preserves the encoder–decoder structure of LM-Net while replacing the original Global Fusion Transformer (GFT) and Local Fusion Transformer (LFT) with the proposed Lightweight Global Bottleneck (LGB) and Lightweight Neighborhood Attention Block (LNAB), respectively. LGB compresses bottleneck features before global self-attention, whereas LNAB applies neighborhood attention to only part of the skip-feature channels.

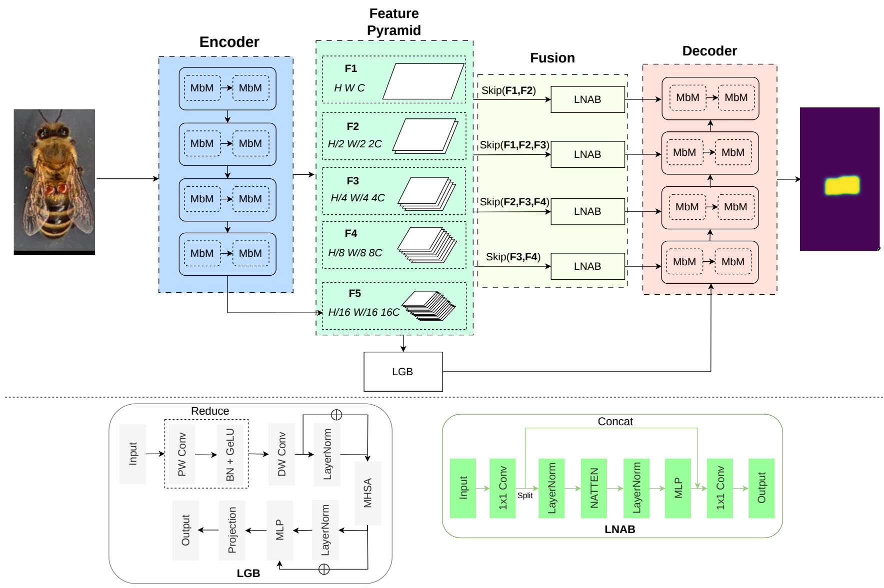

Fig. 2: Overview of the CLM-Net architecture

The encoder consists of an initial stem followed by four hierarchical stages constructed from re-parameterizable convolutional blocks, denoted as Multi-Branch Modules (MbM) in Fig. 2. Stride-2 convolutional downsampling progressively produces five hierarchical feature maps (F_1–F_5). The multi-scale encoder features are first aggregated through the Pyramid Pooling Module and subsequently forwarded to LGB. Let $\mathbf{X} \in \mathbb{R}^{B \times C \times H_b \times W_b}$ denote the aggregated bottleneck feature. LGB reduces its channel dimension from C to d using a 1×1 pointwise convolution, followed by a 3×3 depthwise convolution for local spatial refinement:

$$\mathbf{X}_r = \phi_{\mathrm{dw}}\left(\phi_{\mathrm{pw}}(\mathbf{X})\right), \qquad \mathbf{X}_r \in \mathbb{R}^{B \times d \times H_b \times W_b} \tag{1}$$

The reduced feature is flattened into $\mathbf{Z}_0 \in \mathbb{R}^{B \times N \times d}$, where $N = H_b W_b$, and processed by a Multi-Head Self-Attention (MHSA) module [23] followed by an MLP:

$$\mathbf{Z}_1 = \mathbf{Z}_0 + \mathrm{MHSA}\left(\mathrm{LayerNorm}(\mathbf{Z}_0)\right) \tag{2}$$

$$\mathbf{Z}_2 = \mathbf{Z}_1 + \mathrm{MLP}\left(\mathrm{LayerNorm}(\mathbf{Z}_1)\right) \tag{3}$$

The output sequence is reshaped and projected by a 1×1 convolution to match the decoder channel dimension. Since self-attention is performed over the compressed width, LGB reduces the computational overhead while preserving global spatial interactions.

The decoder contains four progressive upsampling stages. At each stage, the decoder feature is bilinearly upsampled, adjusted by a 3×3 convolution, fused with the corresponding LNAB-refined skip feature by element-wise addition, and processed by re-parameterizable convolutional blocks (MbM). A final 1×1 convolution produces the segmentation logits.

For skip refinement, the multi-scale skip features aggregated by the inherited M2Skip and M3Skip modules of the original LM-Net are subsequently processed by LNAB. Given $\mathbf{S} \in \mathbb{R}^{B \times C_s \times H_s \times W_s}$, LNAB first projects and splits it into an attention branch $\mathbf{S}_a$ and an identity branch $\mathbf{S}_i$:

$$[\mathbf{S}_a, \mathbf{S}_i] = \mathrm{Split}\left(\psi_{\mathrm{in}}(\mathbf{S})\right) \tag{4}$$

Neighborhood Attention [24] and an MLP are applied only to $\mathbf{S}_a$:

$$\mathbf{S}'_a = \mathbf{S}_a + \mathrm{NATTEN}\left(\mathrm{LayerNorm}(\mathbf{S}_a)\right) \tag{5}$$

$$\mathbf{S}''_a = \mathbf{S}'_a + \mathrm{MLP}\left(\mathrm{LayerNorm}(\mathbf{S}'_a)\right) \tag{6}$$

The refined branch is concatenated with the unchanged identity branch and fused by a 1×1 convolution:

$$\mathbf{Y}_{\mathrm{LNAB}} = \psi_{\mathrm{out}}\left(\mathrm{Concat}[\mathbf{S}''_a, \mathbf{S}_i]\right). \tag{7}$$

Each branch contains 50% of the projected channels. The identity branch retains part of the original feature information, while $\mathbf{S}''_a$ provides locally refined features. This partial-attention design reduces computation while limiting the dilution of the original skip information before decoder fusion.

4 Experiments and Results

4.1 Experimental Setup

All experiments were implemented in PyTorch and conducted on an NVIDIA RTX PRO 6000 Blackwell GPU. Each configuration was trained and evaluated over five independent runs with different random seeds, and results are reported as mean $\pm$ standard deviation.

The models were trained using AdamW with an initial learning rate of (3×10^{-4}), a weight decay of (1×10^{-4}), and a batch size of 8. Data augmentation included random horizontal and vertical flipping and random brightness–contrast adjustment. The loss function combined Cross-Entropy and Dice losses with equal weights. Training lasted up to 100 epochs, with early stopping after 10 epochs without improvement in validation loss.

In CLM-Net, LNAB applies neighborhood attention to 50% of the channels using a (3×3) kernel, while LGB uses four attention heads and an MLP expansion ratio of 2. LGB bottleneck widths of 64, 128, 192, 256, and 384 were evaluated.

Performance was measured using Dice and IoU. During inference, foreground masks were obtained by applying softmax to the model outputs and thresholding the foreground probability at 0.7.

4.2 Quantitative Results & Ablation Analysis

Table 2, together with Fig. 3 evaluates the effectiveness of the proposed architectural modifications from both segmentation performance and computational efficiency perspectives. Here, CLM-Net-d denotes the proposed architecture with a bottleneck width of d channels in the Lightweight Global Bottleneck (LGB).

Table 2: Ablation study of different CLM-Net configurations. Best test results are shown in bold

Configuration	Test Dice	Test IoU	Val Dice	Val IoU
CLM-Net-64	69.70 ± 0.69	53.49 ± 0.80	**73.25 ± 0.23**	**57.79 ± 0.29**
CLM-Net-128	70.24 ± 0.57	54.13 ± 0.68	73.02 ± 0.87	57.37 ± 1.13
CLM-Net-192	**70.32 ± 0.90**	**54.24 ± 1.07**	73.01 ± 1.02	57.50 ± 1.26
CLM-Net-256	69.52 ± 1.49	53.30 ± 1.74	72.43 ± 1.21	56.79 ± 1.47
CLM-Net-384	69.71 ± 0.80	53.51 ± 0.94	72.64 ± 0.35	57.03 ± 0.43
LM-Net + LNAB	69.65 ± 0.54	53.43 ± 0.63	72.93 ± 0.82	57.40 ± 1.02
LM-Net	69.41 ± 1.17	53.16 ± 1.36	72.25 ± 0.57	56.56 ± 0.70

Replacing the original Local Fusion Transformer (LFT) with the proposed Lightweight Neighborhood Attention Block (LNAB) results in a higher mean test Dice score of 69.65% and IoU of 53.43%, compared with 69.41% and 53.16% for LM-Net. At the same time, the model complexity is reduced from 4.15M to 3.98M parameters and from 9.899 to 8.796 GFLOPs. These results indicate that applying neighborhood attention to only a subset of feature channels effectively removes redundant feature interactions while preserving discriminative local representations, leading to a more efficient skip refinement strategy.

The influence of the bottleneck projection dimension is further investigated by varying the compressed feature width from 64 to 384 channels. As shown in Table 2, the segmentation performance gradually improves as the bottleneck width increases from 64 to 192 channels, with CLM-Net-192 achieving the highest test Dice score of 70.32% and the highest test IoU of 54.24%. However, further increasing the bottleneck width to 256 and 384 channels does not provide additional accuracy gains and instead slightly degrades the segmentation performance, suggesting that excessively large latent representations introduce redundant feature information without improving discriminative capability.

The computational complexity analysis in Fig. 3 further supports this observation. Increasing the bottleneck width substantially increases both the number of parameters and GFLOPs, whereas the performance improvement quickly saturates. In particular, CLM-Net-192 requires only 1.71M parameters and 8.005 GFLOPs, corresponding to reductions of approximately 58.8% and 19.1%, respectively, compared with the original LM-Net, while simultaneously achieving the best segmentation accuracy. Although CLM-Net-64 further reduces the model size to 1.37M parameters and 7.867 GFLOPs, its segmentation performance is inferior to that of CLM-Net-192. Conversely, larger bottleneck widths such as 256 and 384 increase the computational cost without yielding corresponding accuracy improvements. These results demonstrate that the proposed lightweight bottleneck effectively removes redundant channel representations while preserving global contextual information. Among all evaluated configurations, CLM-Net-192 achieves the best trade-off between segmentation accuracy and computational efficiency, making it the preferred configuration for the proposed architecture.

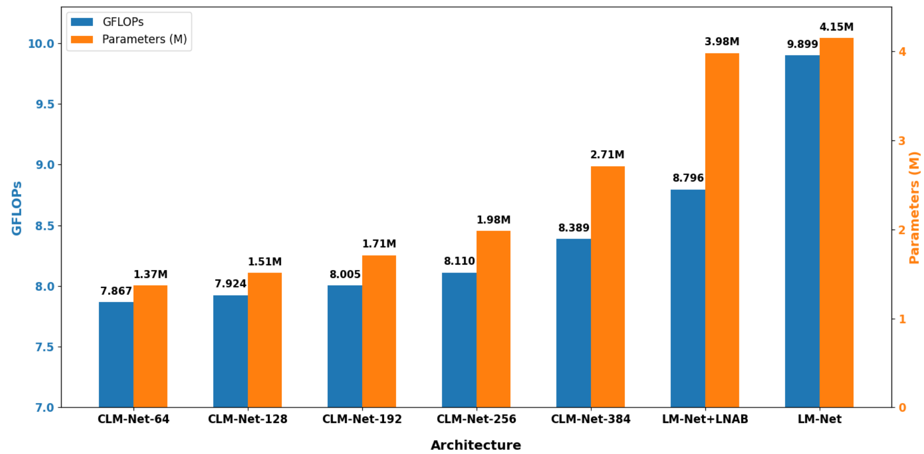

Fig. 3: Comparison of GFLOPs and parameter count across CLM-Net configurations

4.3 Qualitative Visualization Analysis

To further assess the effectiveness of the proposed architecture, qualitative visualization results are presented to compare the segmentation behavior of LM-Net, CLM-Net-128, and CLM-Net-192.

Fig. 4 presents three inference examples. In the first positive sample, all models successfully localize the mite and generate masks that closely match the available reference masks, with CLM-Net-192 achieving the highest Dice and IoU scores. In the negative sample, LM-Net and CLM-Net-128 produce false-positive regions, whereas CLM-Net-192 correctly predicts an empty mask, indicating better suppression of background responses in this case. However, the third example shows a clear limitation, as CLM-Net-192 produces a fragmented prediction compared to the other models. Although these examples are not intended as a comprehensive evaluation, they highlight both the potential advantages and the remaining instability of the proposed architecture across different samples.

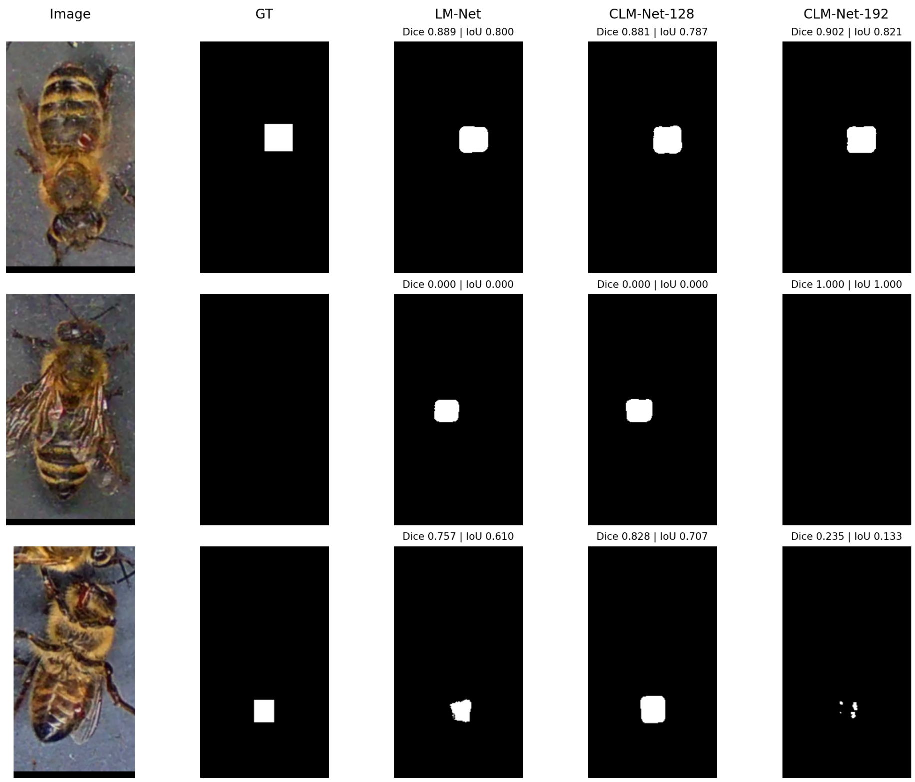

Fig. 4: Qualitative comparison of LM-Net, CLM-Net-128, and CLM-Net-192

4.4 Comparison with the previous study

Figure 5 compares the proposed CLM-Net with previously reported results of existing segmentation models evaluated on the same dataset protocol. The compared methods include UNet, ResUNet, UNet++, UNet3+, DeepLabV3-ResNet101, DeepLabV3-EfficientNetB5, and their corresponding backbone variants [18].

As shown in Fig. 5, CLM-Net achieves a higher Dice score of 70.32% than the previously reported results of the compared methods.. In particular, CLM-Net surpasses the strongest baseline, DeepLabV3-ResNet101, by 3.11% (70.32% vs. 67.21%), indicating competitive segmentation performance compared with the previously reported results.

Compared with conventional encoder–decoder networks, including UNet (61.73%), ResUNet (64.12%), UNet++ (63.03%), and UNet3+ (61.70%), the proposed CLM-Net consistently achieves higher segmentation accuracy. Specifically, CLM-Net improves the Dice score by 8.59%, 6.20%, 7.29%, and 8.62% over UNet, ResUNet, UNet++, and UNet3+, respectively. These results show that CLM-Net provides a clear performance advantage over conventional encoder–decoder segmentation models.

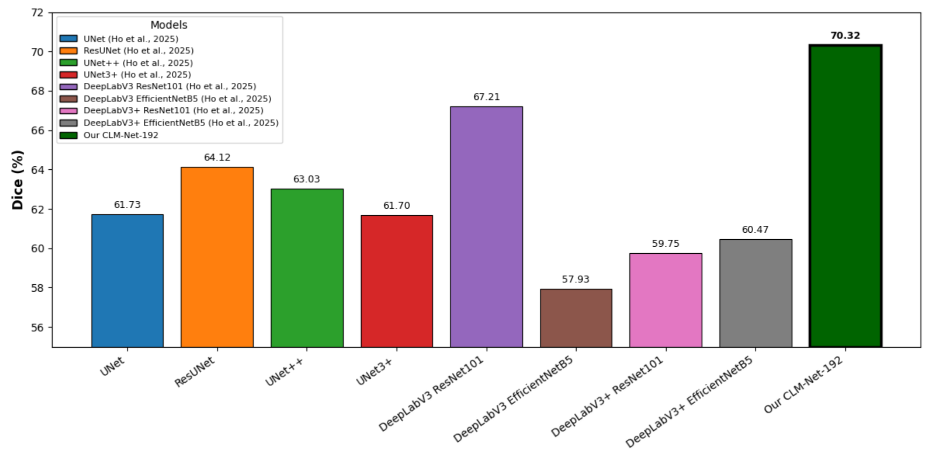

Fig. 5: Comparison of Dice scores between the proposed CLM-Net and existing segmentation models

Moreover, CLM-Net also achieves a higher Dice score than the DeepLabV3-based models with ResNet101 and EfficientNetB5 backbones. Specifically, CLM-Net improves the Dice score by 3.11% over DeepLabV3-ResNet101 and by 12.39% over DeepLabV3-EfficientNetB5. These results show that CLM-Net provides competitive segmentation performance compared with larger architectures. Together with its lightweight parameter design, CLM-Net shows potential for practical Varroa mite segmentation in intelligent beehive monitoring systems.

5 Conclusion and Future Work

In this work, we proposed CLM-Net, a channel-optimized lightweight hybrid CNN–Transformer architecture for Varroa mite segmentation. By introducing the Lightweight Global Bottleneck (LGB) and the Lightweight Neighborhood Attention Block (LNAB), the proposed architecture effectively reduces redundant feature processing while preserving both global contextual information and local structural details. Experimental results on the Bee Varroa dataset demonstrate that CLM-Net achieves an effective balance between segmentation accuracy and computational efficiency. In particular, CLM-Net-192 attains the best performance with a Dice score of 70.32% and an IoU of 54.24%, while reducing the number of parameters from 4.15M to 1.71M and the computational complexity from 9.899 GFLOPs to 8.005 GFLOPs. These results demonstrate that channel-wise optimization provides an effective strategy for maintaining competitive segmentation performance while significantly reducing model complexity. However, the present evaluation is limited to a single dataset, and the generalization of CLM-Net across different imaging conditions and small-object segmentation tasks has not yet been fully established. Future work will therefore investigate adaptive attention mechanisms, bottleneck compression strategies, and cross-dataset evaluation to further improve the efficiency and generalization ability of lightweight segmentation models.

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